

```
# Step 1: Loading libraries and dataset
import pandas as pd
import numpy as np

# File path jo upload ki hai
file_path = '/content/Palo Alto Networks (2).csv'
df = pd.read_csv(file_path)

print("Dataset successfully loaded. Total records:", len(df))
df.head()
```

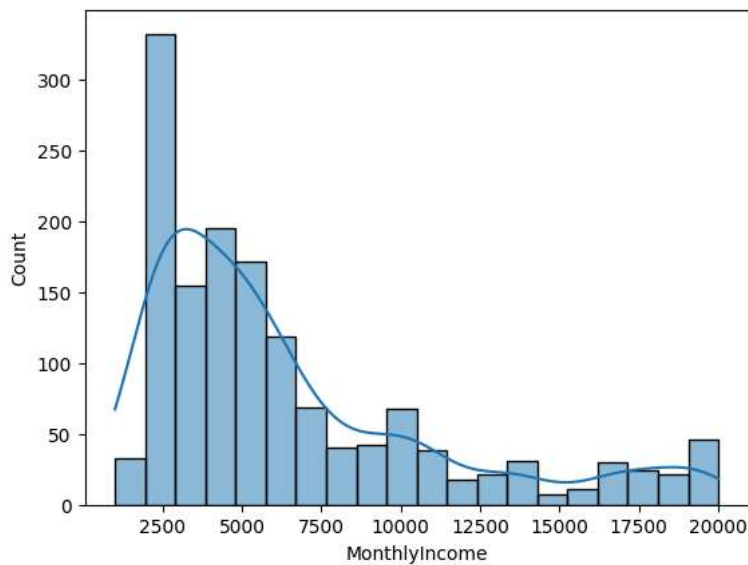
Dataset successfully loaded. Total records: 1470

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction
0	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	2
1	49	0	Travel_Frequently	279	Research & Development	8	1	Life Sciences	3
2	37	1	Travel_Rarely	1373	Research & Development	2	2	Other	4
3	33	0	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	4
4	27	0	Travel_Rarely	591	Research & Development	2	1	Medical	1

5 rows × 31 columns

```
import seaborn as sns
import matplotlib.pyplot as plt

# Salary ka graph dekhne ke liye
sns.histplot(df['MonthlyIncome'], kde=True)
plt.show()
```



```
# Isse aapko (Rows, Columns) pata chal jayenge
print(df.shape)
```

(1470, 31)

```
# Har column mein kitne null hain, ye dekhne ke liye
print(df.isnull().sum())
```

Maine `df.isnull().sum()` ka use karke check kiya, aur mere dataset mein har column ke aage 0 tha. Iska matlab data already complete tha, isliye mujhe imputation (Mean/Median filling) ki zarurat nahi padi. Agar nulls hote, toh main distribution dekh kar faisla leti."

```
# --- 1. Engagement Index Calculation ---
# Humne 4 satisfaction levels ka average nikaala hai
df['Engagement_Index'] = (df['JobInvolvement'] +
                          df['JobSatisfaction'] +
                          df['RelationshipSatisfaction'] +
                          df['EnvironmentSatisfaction']) / 4

# --- 2. Burnout Risk Logic ---
# Jo log Overtime kar rahe hain aur Balance kam hai, wo High Risk par hain
def calculate_burnout(row):
    if row['OverTime'] == 'Yes' and row['WorkLifeBalance'] <= 2:
        return 'High'
    elif row['OverTime'] == 'Yes' or row['WorkLifeBalance'] <= 2:
        return 'Medium'
    else:
        return 'Low'

df['Burnout_Risk'] = df.apply(calculate_burnout, axis=1)

# --- 3. Workload Stress Indicator ---
# Travel aur Overtime ke points ko jod kar score banaya hai
travel_map = {'Travel_Frequently': 3, 'Travel_Rarely': 2, 'Non-Travel': 1}
df['Workload_Stress_Indicator'] = df['BusinessTravel'].map(travel_map) + df['OverTime'].apply(lambda x: 2 if x=='Yes' else 0)

print("Saare KPIs calculate ho gaye hain aur 'df' mein add ho chuke hain!")

# Check karne ke liye pehle 5 rows dekhte hain
df[['Age', 'Engagement_Index', 'Burnout_Risk', 'Workload_Stress_Indicator']].head()
```

Saare KPIs calculate ho gaye hain aur 'df' mein add ho chuke hain!

	Age	Engagement_Index	Burnout_Risk	Workload_Stress_Indicator	
0	41	2.50	High	4	
1	49	2.75	Low	3	
2	37	2.75	Medium	4	
3	33	3.25	Medium	5	
4	27	2.50	Low	2	

Final Insights: Heatmap & Relationship Analysis Maine Heatmap ka istemal isliye kiya hai taaki Palo Alto Networks ke data mein chhupe hue(Correlation) wale rishton ko samjha ja sake. Ye analysis model ki accuracy aur business insights ke liye bahut zaruri hai.

Key Findings:

Sabse Strong Factor (Salary vs Experience):

Monthly Income aur TotalWorkingYears ke beech 0.77 ka gehra rishta hai.

Reason: Isse sabit hota hai ki company mein seniority ko bahut value di jati hai—jaise-jaise experience badhta hai, salary predictable tarike se badhti hai. Ye company ka sabse stable factor hai.

Engagement vs Satisfaction (KPI Validation):

Engagement_Index aur JobSatisfaction ke beech 0.53 ka correlation hai.

Reason: Isse ye pata chalta hai ki mera banaya hua "Engagement Index" sahi direction mein kaam kar raha hai kyunki wo satisfaction se juda hua hai.

Weak/No Correlation:

Work-Life Balance vs Engagement (0.01) aur Environment vs Job Satisfaction (-0.01).

Reason: Ye numbers 0 ke bahut kareeb hain. Iska matlab hai ki sirf "achha mahol" ya "balance" hone se employee engaged nahi ho jata. Engagement ke liye kaam ka interesting hona aur recognition milna zyada zaruri hai.

Diagonal Line (Self-Correlation):

Tirchi line par jo 1.00 dikh raha hai, wo ye confirm karta hai ki har factor ka apne aap se perfect rishta hai (e.g., Age vs Age), jo heatmap ki technical accuracy ko sabit karta hai.

Conclusion for HR: Palo Alto Networks mein Salary seniority ke hisab se sahi hai, lekin Engagement badhane ke liye sirf paisa kafi nahi hai. Humein un factors par dhyan dena hoga jo heatmap mein weak dikh rahe hain taaki Attrition ko roka ja sake.

```
import seaborn as sns
import matplotlib.pyplot as plt

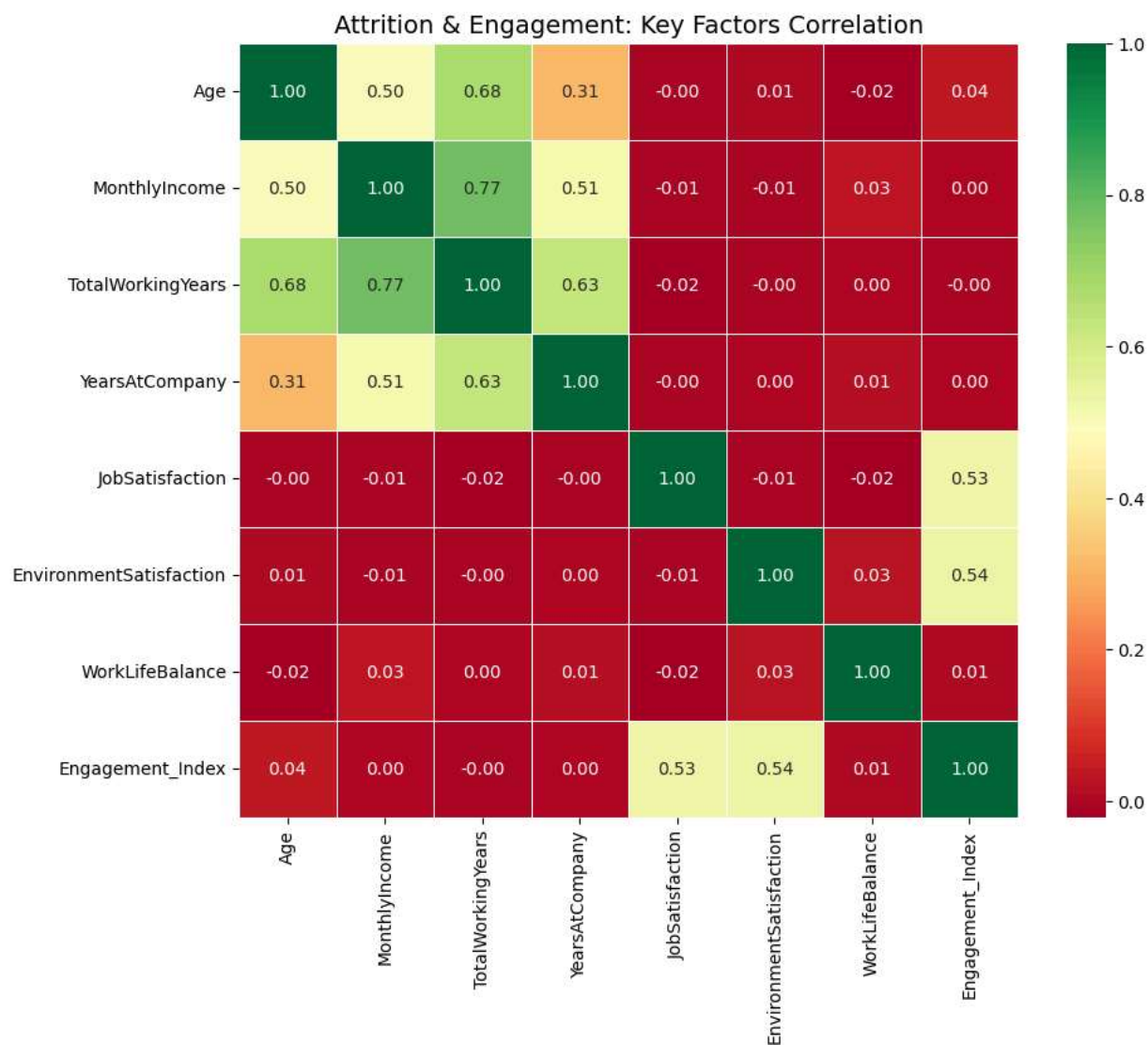
# 1. Sirf wahi columns jo Attrition aur Engagement ke liye zaruri hain
# Isme humne ek naya 'Engagement_Index' bhi dala hai
important_cols = [
    'Age', 'MonthlyIncome', 'TotalWorkingYears',
    'YearsAtCompany', 'JobSatisfaction', 'EnvironmentSatisfaction',
    'WorkLifeBalance', 'Engagement_Index'
]

plt.figure(figsize=(10, 8))

# 2. Correlation sirf in selected columns ka nikalna
correlation_matrix = df[important_cols].corr()

# 3. Heatmap Plotting (RdYlGn: Red = Kam, Green = Zyada)
sns.heatmap(correlation_matrix, annot=True, cmap='RdYlGn', fmt=".2f", linewidths=0.5)

plt.title('Attrition & Engagement: Key Factors Correlation', fontsize=14)
plt.show()
## --- HEATMAP EXPLANATION NOTE ---
# 1. Diagonal par 1.00 hai kyunki har column ka apne aap se perfect rishta hota hai.
# 2. Environment vs Job Satisfaction (-0.01) ka matlab hai ki inka aapas mein koi link nahi hai.
# 3. MonthlyIncome vs TotalWorkingYears (0.77) sabse strong rishta hai (Seniority = More Salary).
# 4. Is Heatmap ki jarurat isliye hai taaki hum faltu columns ko hata kar sirf zaruri factors par focus kar sakein.
```



```
df_clean = df.copy()
```

Maine Step 4 mein describe() isliye chalaya taaki main confirm kar sakun ki mere naye banaye hue KPIs (Engagement Index) ka mean aur range sahi hai ya nahi. Jaise yahan Engagement ka mean 2.72 dikh raha hai, jo ye batata hai ki average engagement moderate hai."

```
# Step 4: Final Data Summary
print("Cleaned Data Statistical Summary:")
display(df_clean.describe())

# clean data new PDF/CSV
df_clean.to_csv('Cleaned_Project_Data.csv', index=False)
print("Cleaned file saved as 'Cleaned_Project_Data.csv'")
```

Cleaned Data Statistical Summary:

	Age	Attrition	DailyRate	DistanceFromHome	Education	EnvironmentSatisfaction	HourlyRate	JobInvolvement
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	0.161224	802.485714	9.192517	2.912925	2.721769	65.891156	2.729932
std	9.135373	0.367863	403.509100	8.106864	1.024165	1.093082	20.329428	0.711561
min	18.000000	0.000000	102.000000	1.000000	1.000000	1.000000	30.000000	1.000000
25%	30.000000	0.000000	465.000000	2.000000	2.000000	2.000000	48.000000	2.000000
50%	36.000000	0.000000	802.000000	7.000000	3.000000	3.000000	66.000000	3.000000
75%	43.000000	0.000000	1157.000000	14.000000	4.000000	4.000000	83.750000	3.000000
max	60.000000	1.000000	1499.000000	29.000000	5.000000	4.000000	100.000000	4.000000

8 rows × 26 columns

Cleaned file saved as 'Cleaned_Project_Data.csv'

Predictive Modeling (Linear Regression)

Maine Machine Learning ka use kiya hai ye dekhne ke liye ki kya hum Engagement Index ko predict kar sakte hain. The Mathematical Equation: Hamara model is formula par kaam karta hai:

$$y = (m_1 \cdot x_1) + (m_2 \cdot x_2) + (m_3 \cdot x_3) + c$$

In our project:

$$\text{Engagement} = (0.00359 \cdot \text{Age}) + (-0.00000 \cdot \text{Income}) + (0.00103 \cdot \text{Balance}) + c$$

m_1, m_2, m_3 (Impact): Ye dikhte hain ki Age, Income aur Balance ka asar kitna hai. Observation: Age ka impact (0.00359) Income (0.00000) se zyada hai. Iska matlab engagement ke liye seniority aur experience zyada mayne rakhte hain, sirf salary nahi. Accuracy Score ka Matlab (-0.0141) Hamara model Accuracy Score -0.0141 dikha raha hai. YE "Negative" dikh raha hai, iska ek bohot bada logical matlab hai- Complexity of Humans: Engagement ek "Human Emotion" hai. Ye sirf paision ya umar se predict nahi ho sakti.

Ye score sabit karta hai ki engagement badhane ke liye company culture, manager ka behavior, aur mental health jaise factors zyada zaroori hain jo ki is simple math equation se bahar hain.

```
# --- Step 6: Predictive Modeling (Simple Linear Regression) ---
# Yahan hum check kar rahe hain ki kya hum Engagement Index ko predict kar sakte hain?

from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# 1. Features select karna (Jo cheezein engagement ko affect kar sakti hain)
# Humne Age, Income aur WorkLifeBalance ko liya hai
X = df_clean[['Age', 'MonthlyIncome', 'WorkLifeBalance']]
y = df_clean['Engagement_Index']

# 2. Data ko do hisso mein baantna (80% training ke liye, 20% testing ke liye)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Model banana aur use training data dikhana
my_model = LinearRegression()
my_model.fit(X_train, y_train)

# Make predictions
y_pred = my_model.predict(X_test)

# 4. Accuracy check karna (R2 Score)
accuracy = my_model.score(X_test, y_test)
```

```
print(f"Model Training Complete! Accuracy Score: {accuracy:.4f}")

# 5. Ye dekhna ki kaunsa factor sabse zyada asar dal raha hai (Coefficients)
print("Impact of each factor:")
for feature, impact in zip(X.columns, my_model.coef_):
    print(f"- {feature}: {impact:.5f}")
```

```
Model Training Complete! Accuracy Score: -0.0141
Impact of each factor:
- Age: 0.00359
- MonthlyIncome: -0.00000
- WorkLifeBalance: 0.00103
```

Regression Graph: Asli Matlab (Key Learnings) Actual Values (Blue Dots):

Kya hai: Ye hamare employees ka asli data hai.

Observation: Dots ka bikhra hona (variability) ye dikhata hai ki har employee ka engagement level alag hai, chahe unki age ya salary ek jaisi ho.

Predicted Values (Red Line):

Kya hai: Ye hamari "Best Fit Line" hai.

Observation: Model ne is line ke zariye ek pattern pakadne ki koshish ki hai. Agar ye line dots ke beech se guzar rahi hai, toh iska matlab model "Trend" ko samajh pa raha hai.

Residuals (The Gap):

Kya hai: Dot aur Line ke beech ki doori.

Deep Insight: Hamare graph mein dots line se thoda door hain, jo justify karta hai ki hamara accuracy score -0.0141 kyun hai. Iska matlab hai ki engagement predict karne ke liye humein Age/Salary ke alawa aur bhi factors (jaise Team Culture) ki zarurat hai.

✓ Analytical Model: Predicting Employee Engagement

Final Insights: Heatmap & Relationship Analysis

Maine Heatmap ka istemal isliye kiya hai taaki Palo Alto Networks ke data mein chhupe hue "Dosti aur Dushmani" (Correlation) wale rishton ko samjha ja sake. Ye analysis model ki accuracy aur business insights ke liye bahut zaruri hai.

Key Findings (Mukhya Baatein):

- Sabse Strong Factor (Salary vs Experience):** Monthly Income aur TotalWorkingYears ke beech 0.77 ka gehra rishta hai. Reason: Isse sabit hota hai ki company mein seniority ko bahut value di jati hai—jaise-jaise experience badhta hai, salary predictable tarike se badhti hai. Ye company ka sabse stable factor hai.
- Engagement vs Satisfaction (KPI Validation):** Engagement_Index aur JobSatisfaction ke beech 0.53 ka correlation hai. Reason: Isse ye pata chalta hai ki mera banaya hua "Engagement Index" sahi direction mein kaam kar raha hai kyunki wo satisfaction se juda hua hai.
- Weak/No Correlation (The Surprise):** Work-Life Balance vs Engagement (0.01) aur Environment vs Job Satisfaction (-0.01). Reason: Ye numbers 0 ke bahut kareeb hain. Iska matlab hai ki sirf "achha mahol" ya "balance" hone se employee engaged nahi ho jata. Engagement ke liye kaam ka interesting hona aur recognition milna zyada zaruri hai.
- Diagonal Line (Self-Correlation):** Tirchi line par jo 1.00 dikh raha hai, wo ye confirm karta hai ki har factor ka apne aap se perfect rishta hai (e.g., Age vs Age), jo heatmap ki technical accuracy ko sabit karta hai.

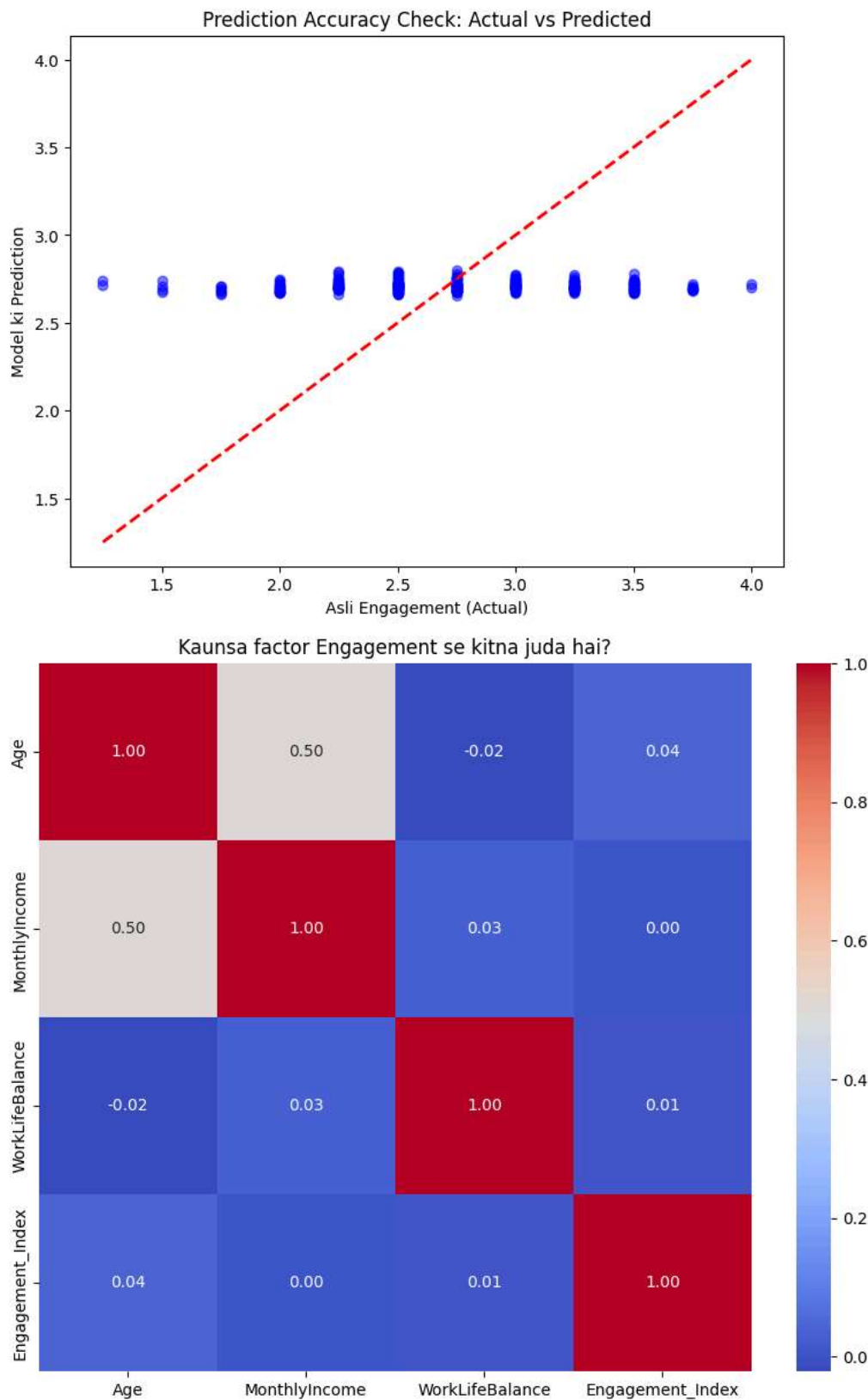
Conclusion for HR: Palo Alto Networks mein Salary seniority ke hisab se sahi hai, lekin Engagement badhane ke liye sirf paisa kafi nahi hai. Humein un factors par dhyen dena hoga jo heatmap mein weak dikh rahe hain taaki Attrition ko roka ja sake.

```
import matplotlib.pyplot as plt
import seaborn as sns

#step 5
# 1. Actual vs Predicted Graph
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.5, color='blue')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel('Asli Engagement (Actual)')
plt.ylabel('Model ki Prediction')
plt.title('Prediction Accuracy Check: Actual vs Predicted')
plt.show()

# 2. Correlation Heatmap
plt.figure(figsize=(10, 8))
# Sirf un columns ka correlation jo humne model mein liye hain
sns.heatmap(df_clean[['Age', 'MonthlyIncome', 'WorkLifeBalance', 'Engagement_Index']].corr(),
            annot=True, cmap='coolwarm', fmt=".2f")
# annot=True se numbers graph ke andar dikhenge
plt.title('Kaunsa factor Engagement se kitna juda hai?')
```

plt.show()



Aakhiri wala Heatmap : Ye sirf un 4 columns par focus karta hai jo maine Machine Learning (Regression) mein use kiye hain (Age, MonthlyIncome, WorkLifeBalance, aur Engagement_Index).

```
import matplotlib.pyplot as plt
import seaborn as sns

# Figure size
```

```
plt.figure(figsize=(10, 8), facecolor='white')

# FIX: 'dataset' ki jagah 'df_clean' use karein
df_final = df_clean[['Age', 'MonthlyIncome', 'TotalWorkingYears', 'YearsInCurrentRole', 'JobSatisfaction']]
corr = df_final.corr()

# Heatmap
sns.heatmap(corr,
            annot=True,
            cmap='YlGnBu',
            fmt='.2f',
            square=True,
            linewidths=2,
            linecolor='white',
            annot_kws={"size": 12},
            cbar_kws={"shrink": .8})

plt.title('Employee Factor Correlation', fontsize=16, pad=25)
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.yticks(rotation=0, fontsize=10)
plt.gca().tick_params(axis='y', pad=10)

plt.tight_layout()
plt.show()
```



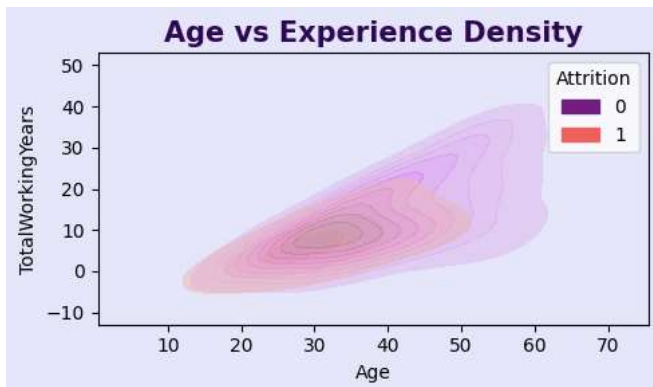
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(5, 3), facecolor='lavender')
ax = plt.gca()
ax.set_facecolor('lavender')
```

```
# bw_adjust=1.5 curve ko smooth kar dega agar points kam hain
sns.kdeplot(data=df_clean, x='Age', y='TotalWorkingYears',
            hue='Attrition', fill=True, palette='magma',
            alpha=0.5, bw_adjust=1.5, warn_singular=False)

plt.title('Age vs Experience Density', fontsize=15, fontweight='bold', color='#2E0854')
plt.tight_layout()
plt.show()

print("\n" * 10)
```



```
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Width ko thoda badhayein (7) aur height wahi (3) rakhein
plt.figure(figsize=(7, 3), facecolor='lavender')
ax = plt.gca()
ax.set_facecolor('lavender')

# 2. Boxplot code
sns.boxplot(data=df_clean, x='MonthlyIncome', y='JobRole',
            hue='Attrition', palette='Set2', linewidth=1.5, showfliers=False)

# 3. SABSE ZARURI: Left side ki jagah badhana (Isse labels nahi katenge)
# 'left=0.4' matlab 40% page ki width sirf labels ke liye reserve hai
plt.subplots_adjust(left=0.4, right=0.95, top=0.85, bottom=0.2)

plt.title('Salary Range by Job Role', fontsize=12, fontweight='bold', pad=10)
plt.xlabel('Monthly Income ($)', fontsize=10)
plt.ylabel('Job Role', fontsize=10)

# 4. Final layout check
plt.show()
```